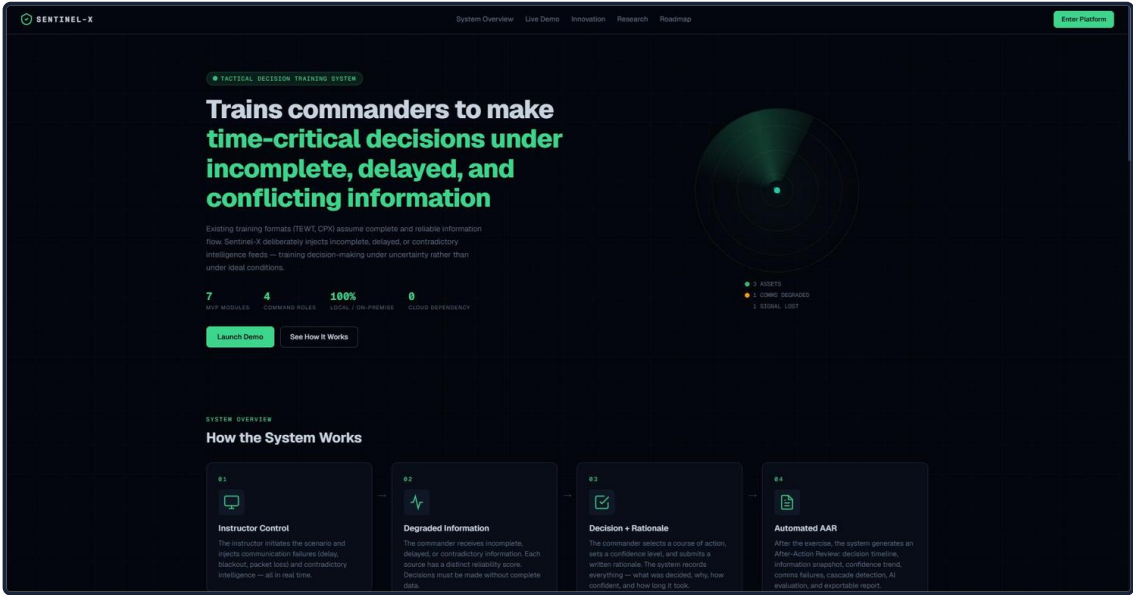


SENTINEL-X

Multi-Domain Decision Trainer for Degraded Communication Environments

USER MANUAL

Version 5.0 | Fullstack Edition



Landing Page — Home

Landing Page — Home

Top Navigation Bar

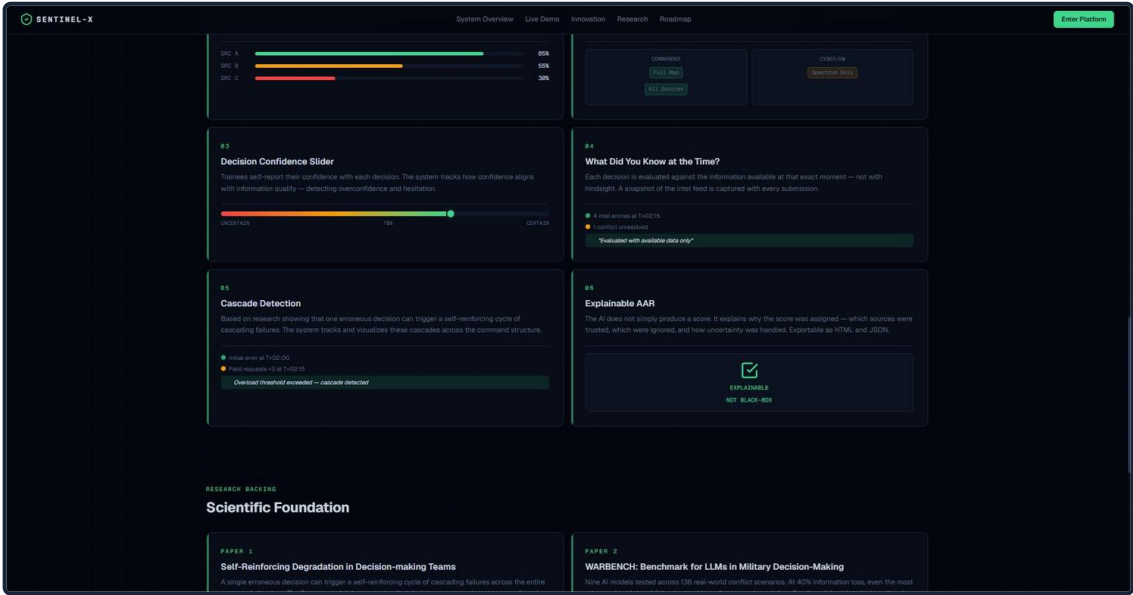
SENTINEL-X logo on the left. Navigation links: System Overview, Live Demo, Innovation, Research, Roadmap. 'Enter Platform' button on the right — click this to access the login/register page.

Hero Section

Shows the main headline: 'Trains commanders to make time-critical decisions under incomplete, delayed, and conflicting information'. Key stats: 7 MVP Modules, 4 Command Roles, 100% Local/On-Premise, 0 Cloud Dependency. Two buttons: 'Launch Demo' and 'See How It Works'.

Tactical Radar Visual

Right side shows a dark circular tactical radar display with active asset status indicators (3 Assets, 1 Comms Degraded, 1 Signal Lost).



Landing Page — Innovation Section

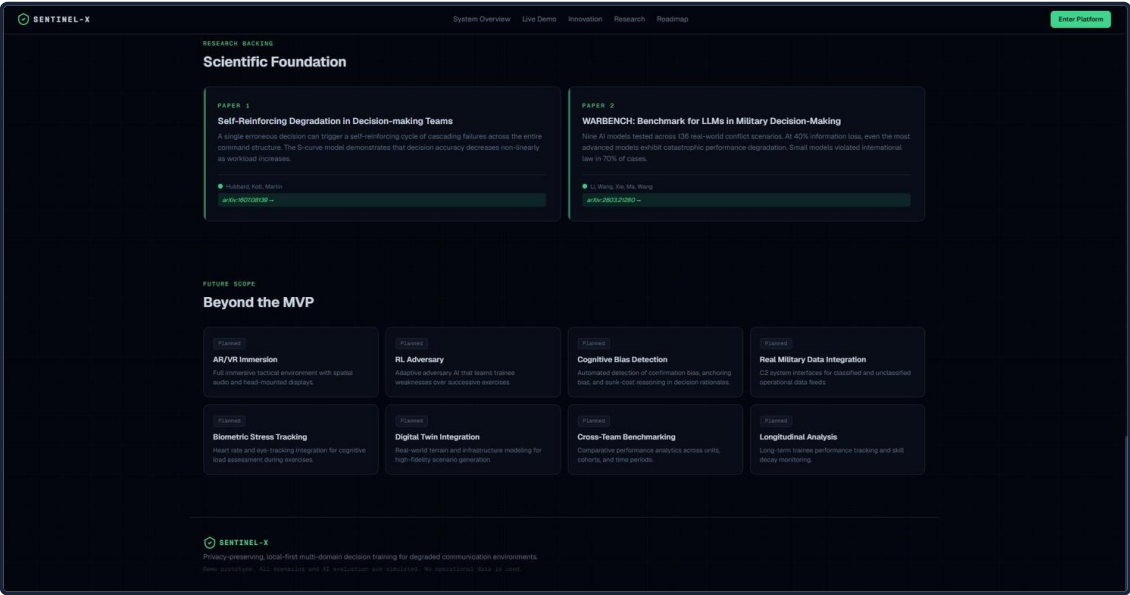
Landing Page — Innovation Section

How the System Works

Four sequential process cards: 01 Instructor Control (injects real-time communication failures), 02 Degraded Information (commander receives incomplete/conflicting data), 03 Decision + Rationale (course of action selection, confidence levels, rationales recorded), 04 Automated AAR (generates post-exercise After-Action Review and analytics).

Live Demo Section

Interactive demo tabs showing: Normal Comms, Comms Degraded, Conflicting Intel, Decision, and AAR phases. 'Launch Full Simulation' button at the bottom.



Landing Page — Research & Roadmap

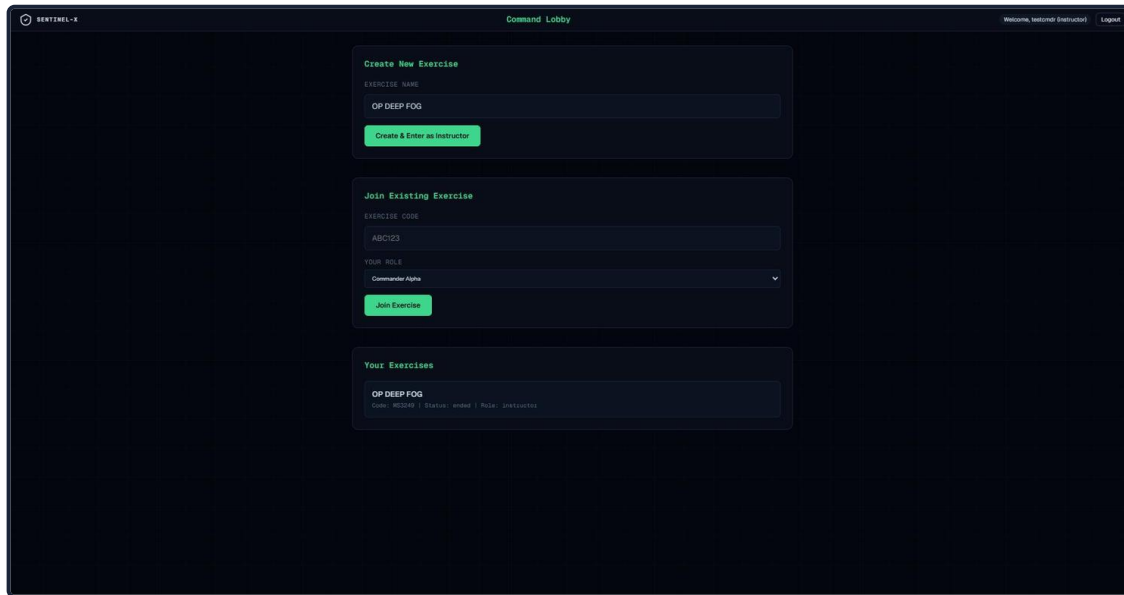
Landing Page — Research & Roadmap

Research Backing

Academic papers that inform the system design. Links to arXiv:1607.08139 (self-reinforcing degradation in decision-making teams by Hubbard, Kott, Martin) and arXiv:2603.21280. These papers form the theoretical basis for cascade detection and degraded communication training.

Roadmap Section

Shows the development roadmap with future milestones and planned features for the Sentinel-X platform.

*Command Lobby — Create & Join Exercises*

Command Lobby — Create & Join Exercises

Header

Shows 'Welcome, testcmdr (Instructor)' with a Logout button. The Command Lobby title is displayed in green.

Create New Exercise

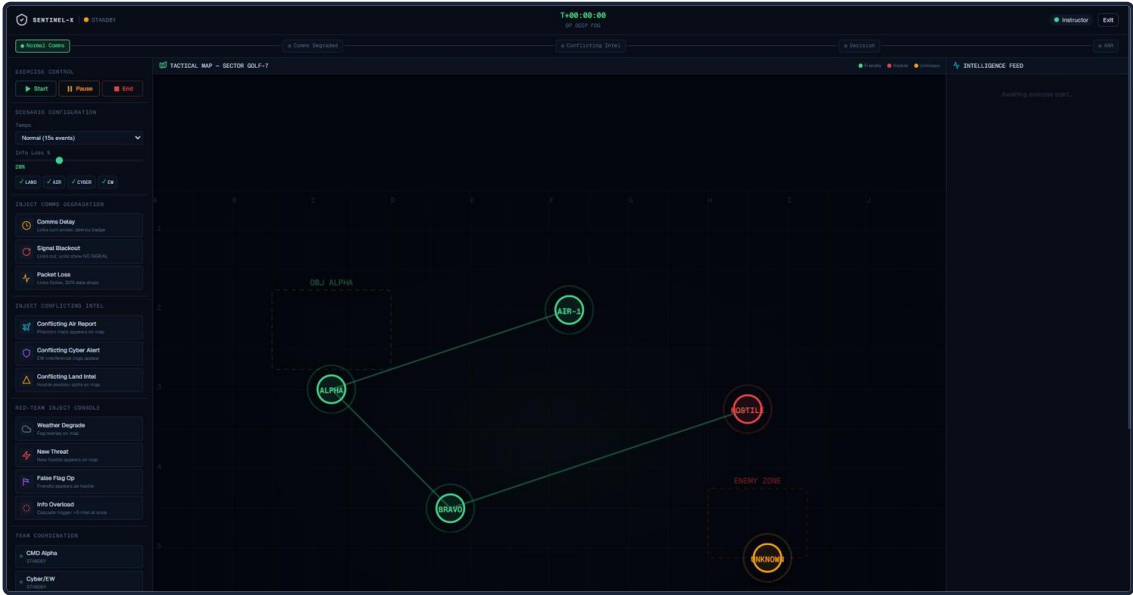
Enter an exercise name (e.g., OP DEEP FOG) in the EXERCISE NAME field and click 'Create & Enter as Instructor'. The system generates a unique code and you enter the simulator as instructor.

Join Existing Exercise

Enter the exercise code provided by your instructor, select your role (Commander Alpha, Cyber/EW Officer, Air Liaison, or Intel Analyst) from the dropdown, and click 'Join Exercise'.

Your Exercises

Lists all exercises you have created or joined, showing name, code, status (created/active/ended), and your role.



Simulator — Standby Mode (Before Start)

Simulator — Standby Mode (Before Start)

Top Status Bar

SENTINEL-X logo with STANDBY status badge. Exercise timer shows T+00:00:00. Exercise name (OP DEEP FOG) displayed. Role indicator shows 'Instructor'. Exit button on the right.

Phase Tabs

Five tabs showing exercise progression: Normal Comms (active), Comms Degraded, Conflicting Intel, Decision, and AAR.

Left Sidebar — Exercise Control

Start, Pause, and End buttons. Scenario Configuration: Tempo dropdown (Slow/Normal/Fast), Info Loss % slider, Domain checkboxes (Land, Air, Cyber, EW).

Left Sidebar — Inject Controls

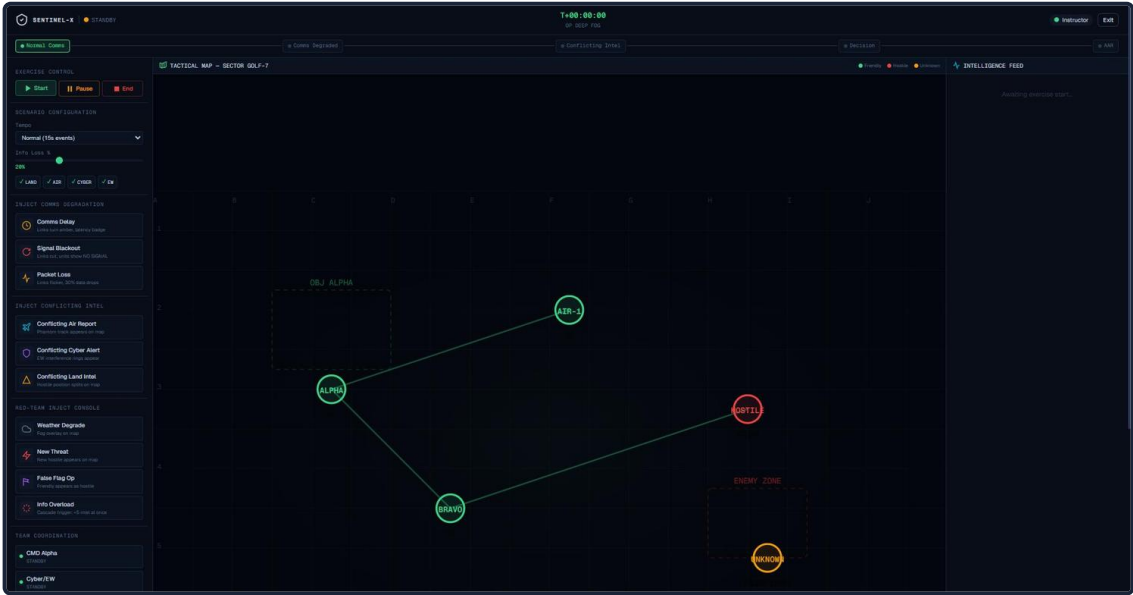
Three groups: Inject Comms Degradation (Comms Delay, Signal Blackout, Packet Loss), Inject Conflicting Intel (Air Report, Cyber Alert, Land Intel), Red-Team Inject Console (Weather Degrade, New Threat, False Flag Op, Info Overload).

Center — Tactical Map

Grid map of Sector Golf-7 with coordinate labels A-J and 1-5. Shows friendly units (ALPHA, BRAVO, AIR-1 in green), hostile units (HOSTILE in red), unknown units (UNKNOWN, UNCONFIRMED in yellow), objective zones (OBJ ALPHA, ENEMY ZONE).

Right — Intelligence Feed

Displays 'Awaiting exercise start...' until the exercise begins.



Simulator — Exercise Active (Normal Comms)

Simulator — Exercise Active (Normal Comms)

Status Change

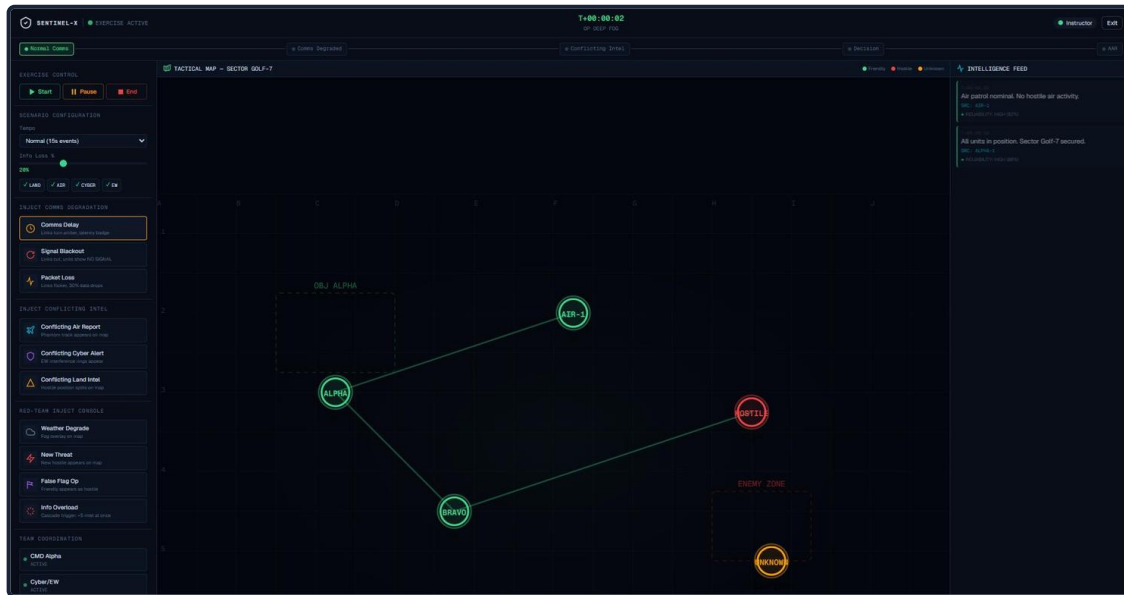
Status badge changes from STANDBY to EXERCISE ACTIVE (green). Timer starts counting. Team coordination roles (CMD Alpha, Cyber/EW) change to ACTIVE.

Intel Feed Populated

The Intelligence Feed starts receiving baseline reports: 'Air patrol nominal', 'All units in position'. Each entry shows timestamp, message, source, and reliability score (HIGH/MEDIUM/LOW).

Map State

Tactical map shows normal green communication links between units. All friendly units (ALPHA, BRAVO, AIR-1) are connected and operational.



Simulator — Comms Degradation Injected

Simulator — Comms Degradation Injected

Comms Delay Effect

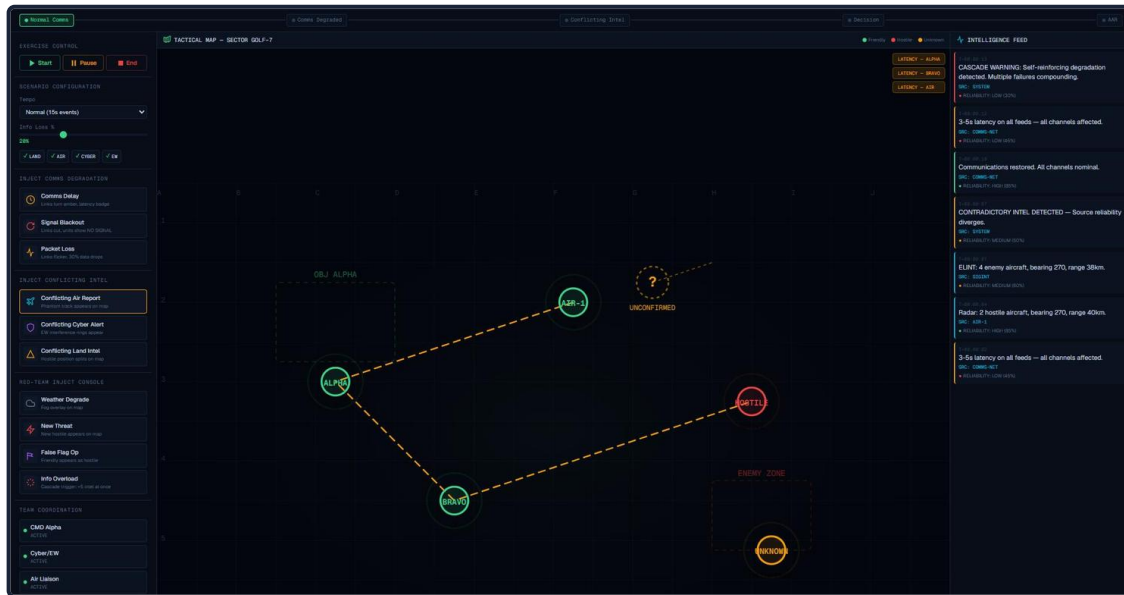
The instructor clicked 'Comms Delay'. Network links on the tactical map turn into dashed amber lines. Yellow latency badges (LATENCY - ALPHA, LATENCY - BRAVO, LATENCY - AIR) appear on the map.

Intel Feed Warning

The Intelligence Feed logs: '3-5s latency on all feeds — all channels affected'. This simulates real-world communication delays in degraded environments.

Auto-Recovery

After approximately 8 seconds, the degradation automatically recovers. A 'COMMS RESTORED' message is injected into the intel feed, and links return to green.



Simulator — Conflicting Intel & Cascade Warning

Simulator — Conflicting Intel & Cascade Warning

Phantom Contact

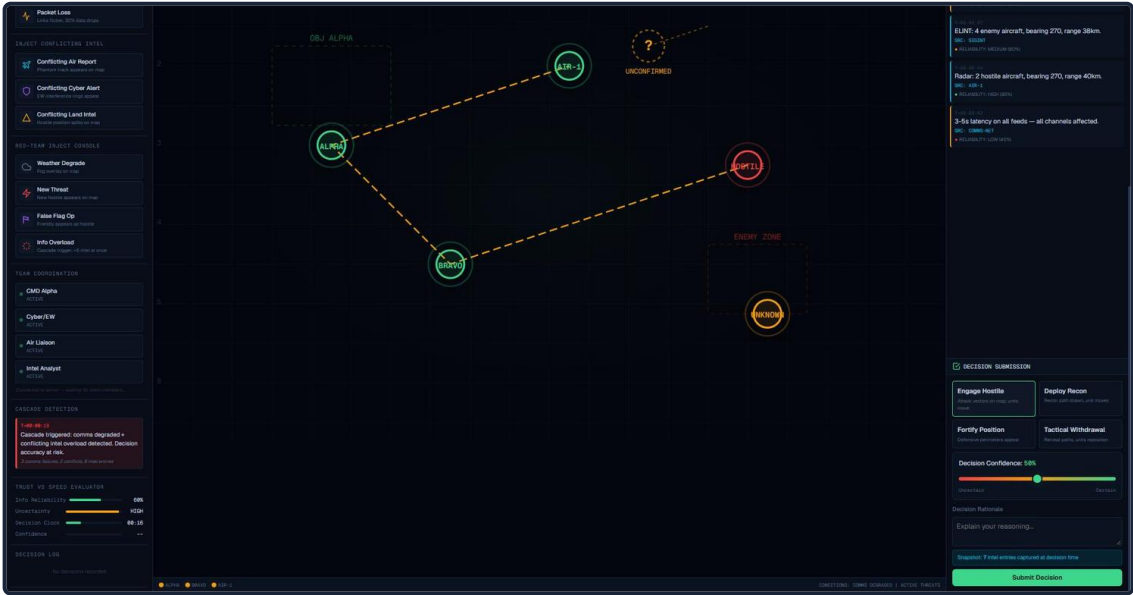
After injecting 'Conflicting Air Report', a dashed unconfirmed phantom contact marked '? UNCONFIRMED' appears near AIR-1 on the tactical map. This represents divergent sensor data.

Diverging Intel

The Intelligence Feed shows contradictory reports: Radar reports 2 hostile aircraft, while ELINT reports 4 enemy aircraft. The reliability scores differ, forcing the commander to assess which source to trust.

Cascade Warning

When 2+ comms channels are degraded AND 1+ conflicting intel is present, the system automatically detects a cascade condition and injects a 'CASCADE WARNING: Self-reinforcing degradation detected' into the intel feed.



Decision Submission & Confidence Slider

Decision Submission & Confidence Slider

Decision Options

Four course-of-action buttons: Engage Hostile (attack vectors drawn, units move to engage), Deploy Recon (recon path drawn, unit moves to investigate), Fortify Position (defensive perimeters appear), Tactical Withdrawal (retreat paths drawn, units reposition).

Confidence Slider

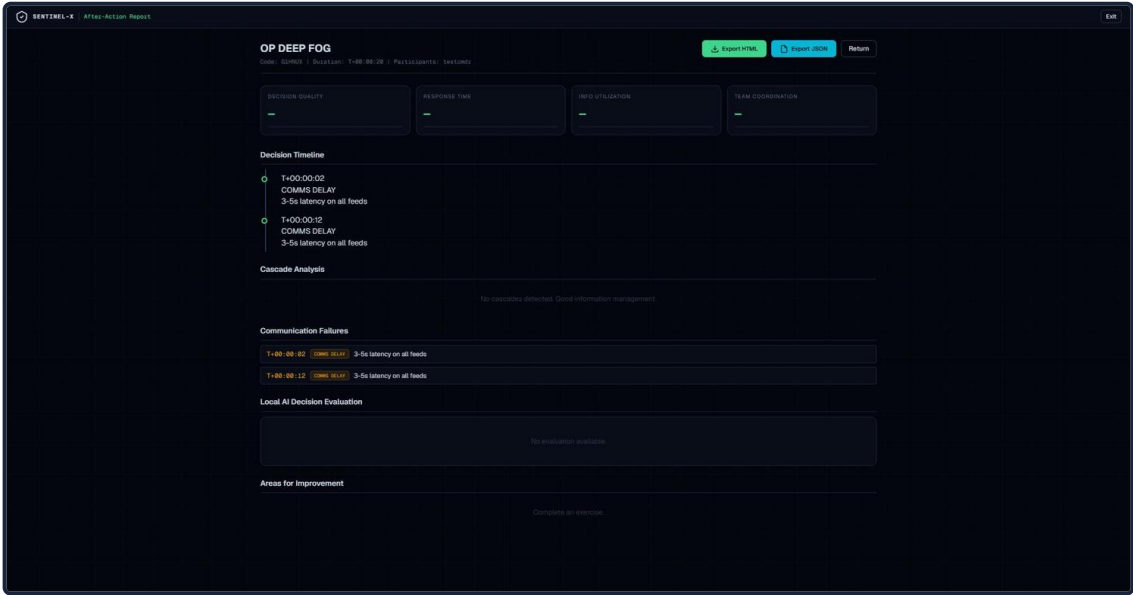
After selecting a decision, a confidence slider appears (0-100%). The trainee self-reports their confidence level. The system tracks overconfidence (high confidence with limited intel) and hesitation (low confidence with ample intel).

Rationale Input

A text area for the trainee to provide written rationale for their decision. This is captured along with an intel snapshot at the moment of submission — supporting the 'What Did You Know at the Time?' evaluation.

Cascade Detection Panel

Shows real-time cascade status. Trust vs Speed Evaluator displays Info Reliability %, Uncertainty level, and Decision Clock.



After-Action Review (AAR) — Timeline & Analysis

After-Action Review (AAR) — Timeline & Analysis

AAR Header

Shows exercise name (OP DEEP FOG), code, duration (T+00:00:20), and participants. Export HTML, Export JSON, and Return buttons in the top right.

Score Cards

Four metric cards: Decision Quality, Response Time, Info Utilization, Team Coordination — each scored 0-100 based on exercise performance.

Decision Timeline

Chronological view of all events: comms delays, conflicting intel, decisions made, and cascades detected. Each event shows timestamp and description.

Cascade Analysis

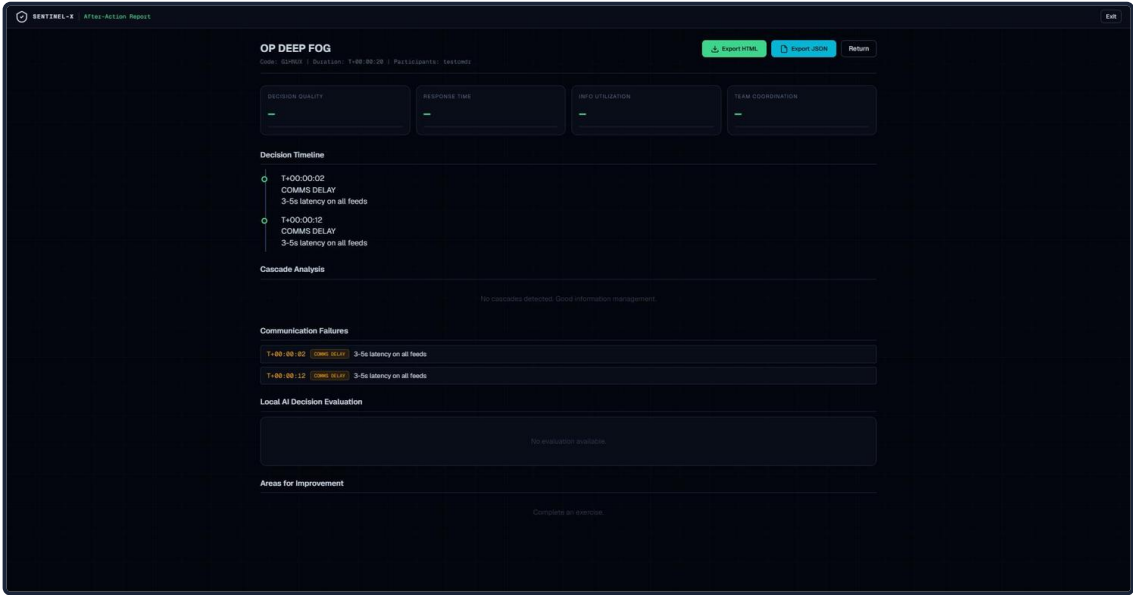
Details any detected cascades with trigger descriptions. Shows 'No cascades detected' if information was managed well.

Communication Failures Log

All injected degradation events with timestamps — comms delays, blackouts, packet losses.

AI Evaluation & Improvement

Local AI evaluation gives overall rating (EXCELLENT/PROFICIENT/DEVELOPING/NEEDS RETRAINING) with findings and personalized recommendations.



AAR — Export Options

AAR — Export Options

Export HTML

Click 'Export HTML' to download a fully formatted report with tables, timelines, and score cards. This creates a printable, shareable document.

Export JSON

Click 'Export JSON' to download raw structured data. Useful for integration with other systems, programmatic analysis, or archival.

Return Button

Click 'Return' to go back to the Command Lobby where you can create a new exercise or join another one.